

1. Can a correct visualization still mislead?

- Medical evidence: the mathematics behind the chart
- AI-854 · Lecture 5
- Cholesterol, HDL and added sugar

2. Three pitches for a health editor

- A: “Cholesterol is dangerous.”
- B: “Raising HDL protects the heart.”
- C: “Sugar is the culprit, so cholesterol was wrongly blamed.”
- For each pitch: initial verdict, missing evidence, and a figure you would need.

3. Funding and selected evidence

- A historical analysis documents an industry-funded review that emphasized fat and cholesterol and downplayed sucrose evidence.
- The sponsor helped shape the review and received drafts. The review omitted its funding disclosure.
- This motivates an evidence audit. The medical claims still need testing.

4. The measurements in the story

Quantity	Unit or encoding
LDL-C / HDL-C	mg/dL
Cardiovascular endpoint	0 or 1 per participant
Added sugar	% of total energy
Published HR	Dimensionless ratio

- LDL-C and HDL-C measure cholesterol carried in different lipoprotein fractions.
- Total serum cholesterol and blood glucose are distinct laboratory measurements.
- Added sugar is a dietary exposure, expressed here as a percentage of energy.
- A biomarker change and a patient outcome answer different questions.

5. Study design sets the claim’s scope

Evidence	Design	Classroom quantity
HPS	Randomized trial	Risk difference and RR
AIM-HIGH	Randomized add-on trial	RR and published HR
NHANES	Observational cohort	Published adjusted HRs

- Randomization supports a comparison of assigned treatments.
- A cohort model reports an association under its adjustment assumptions.
- Endpoint, population and follow-up must accompany the estimate.

6. HPS: participants and first events

Assigned arm	Participants n	With major vascular event e
Simvastatin	10,269	2,033
Placebo	10,267	2,585

- Composite outcome: major coronary event, stroke or revascularization.
- Each affected participant contributes once to this composite. Component counts can overlap.

7. Binary outcomes estimate event risk

- $Y_i = 1$ if participant i experiences the selected event, otherwise 0
- $\hat{p} = (1/n) \sum_i Y_i = e/n$
- $E(Y_i) = p$ $\text{Var}(Y_i) = p(1 - p)$
- $E(\hat{p}) = p$

8. Variance of an estimated risk

- $\text{Var}(\hat{p}) = \text{Var}(\sum_i Y_i)/n^2$
- Independence removes the covariance terms
- $\text{Var}(\hat{p}) = np(1 - p)/n^2 = p(1 - p)/n$
- Estimated $\text{SE}(\hat{p}) = \sqrt{[\hat{p}(1 - \hat{p})/n]}$

9. Absolute and relative effects

- $ARR = \hat{p}_C - \hat{p}_T$
- $RR = \hat{p}_T / \hat{p}_C$ $RRR = 1 - RR$
- $1000 \times ARR$ estimates fewer affected people per 1,000
- ARR has probability units. RR and RRR are ratios.

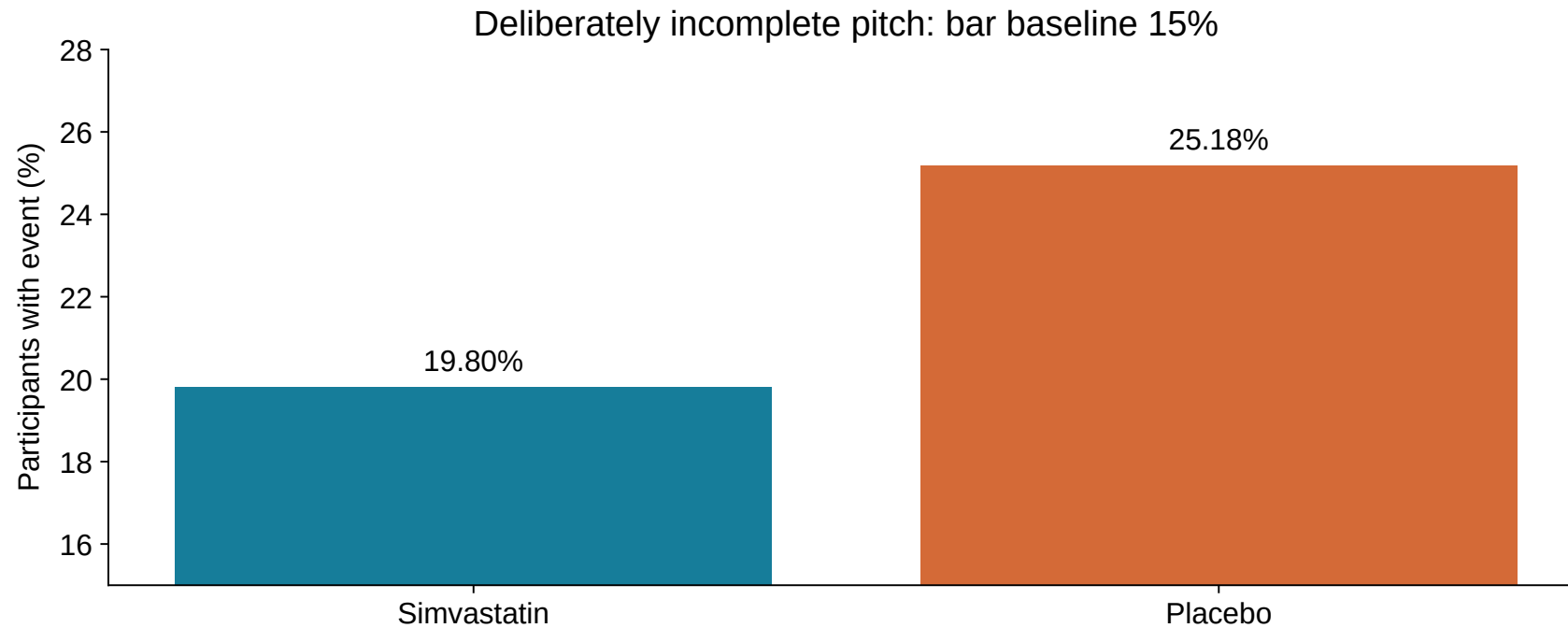
10. HPS: the hand calculation

- Simvastatin: $2,033 / 10,269 = 19.80\%$
- Placebo: $2,585 / 10,267 = 25.18\%$
- ARR = 5.38 percentage points
- RR = 0.786 RRR = 21.4%

11. An interval for absolute benefit

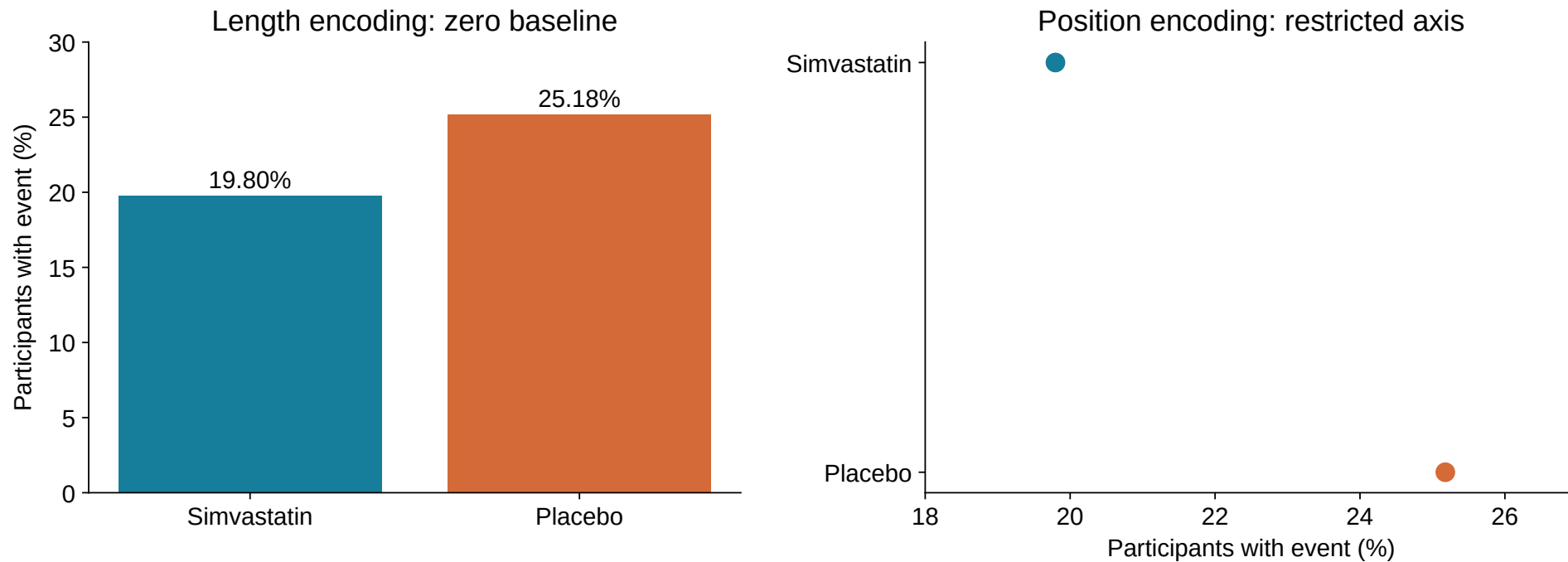
- $SE(ARR) = \sqrt{[\hat{p}_C(1 - \hat{p}_C)/n_C + \hat{p}_T(1 - \hat{p}_T)/n_T]}$
- Approximate 95% CI = $ARR \pm 1.96 \times SE(ARR)$
- HPS: 4.24 to 6.52 percentage points
- About 54 fewer affected people per 1,000 (interval 42–65)

12. A shortened baseline exaggerates length



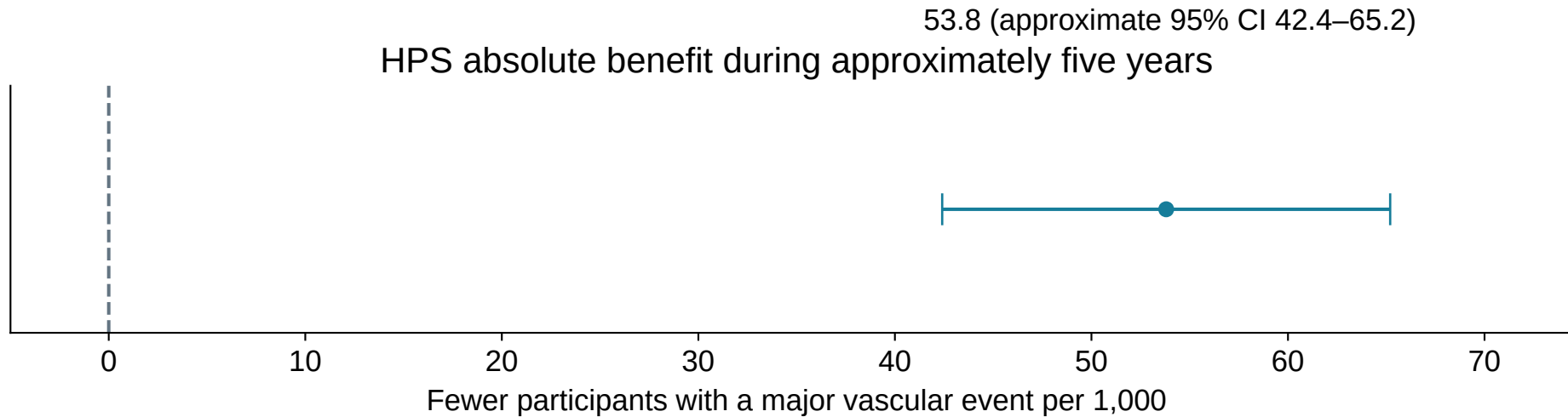
- At a 15% baseline, visible heights have ratio 2.12
- The actual control/treatment risk ratio is 1.27
- Apparent ratio = $(\hat{p}_C - b) / (\hat{p}_T - b)$

13. A repair follows the encoding



- Zero-baseline bars preserve magnitude through length.
- Restricted-axis dots can expose a small difference through position.
- Axis labels and the comparison's units stay visible.

14. Absolute benefit on a common denominator

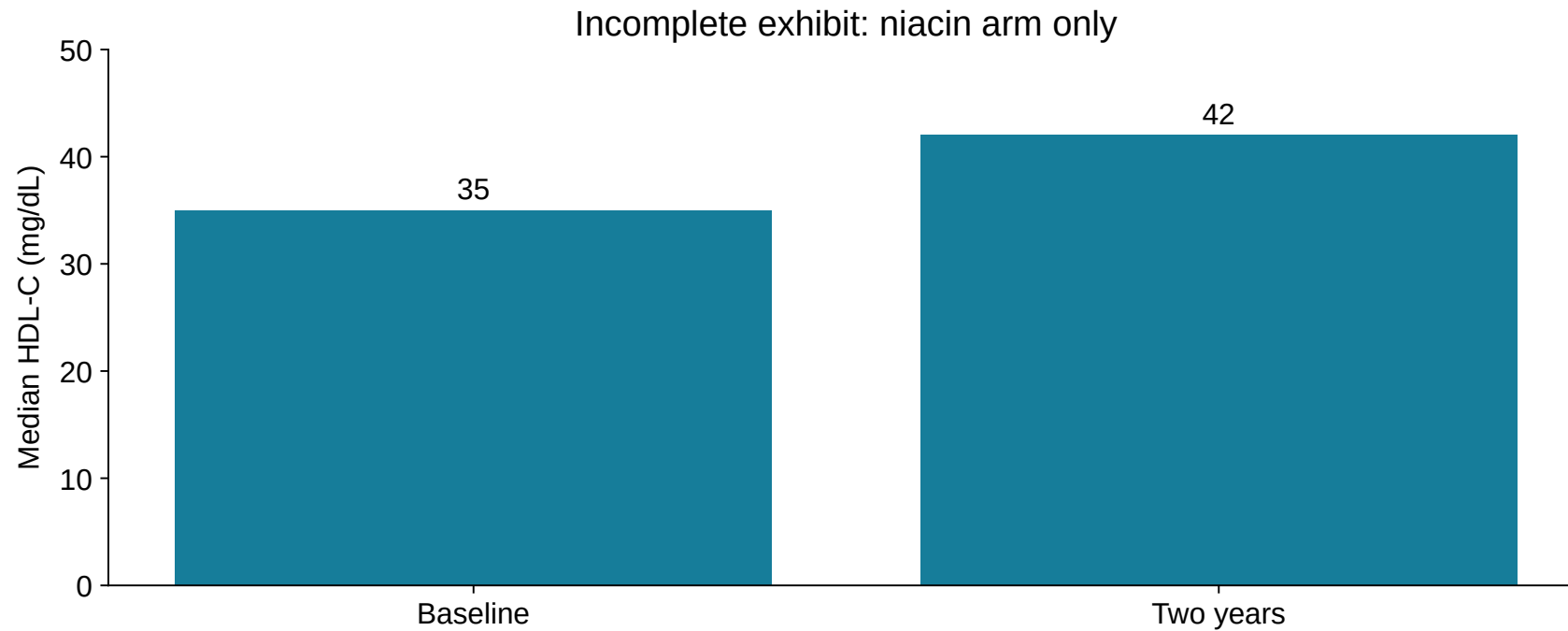


- Approximately 198 versus 252 affected participants per 1,000
- Estimated benefit: 53.8 per 1,000
- The display scales the observed risks to equal denominators.

15. The HPS caption and causal scope

- Assignment to simvastatin reduced major vascular events by 5.38 percentage points during approximately five years.
- The count-based estimate uses affected participants and assigned-arm denominators.
- Wider evidence supports LDL as a causal contributor to atherosclerotic disease.

16. HDL: a biomarker pitch



- Median HDL-C in the niacin arm rose from 35 to 42 mg/dL at two years.
- $(42 - 35) / 35 = 20\%$
- What evidence would connect this rise to protection?

17. AIM-HIGH: the clinical outcome

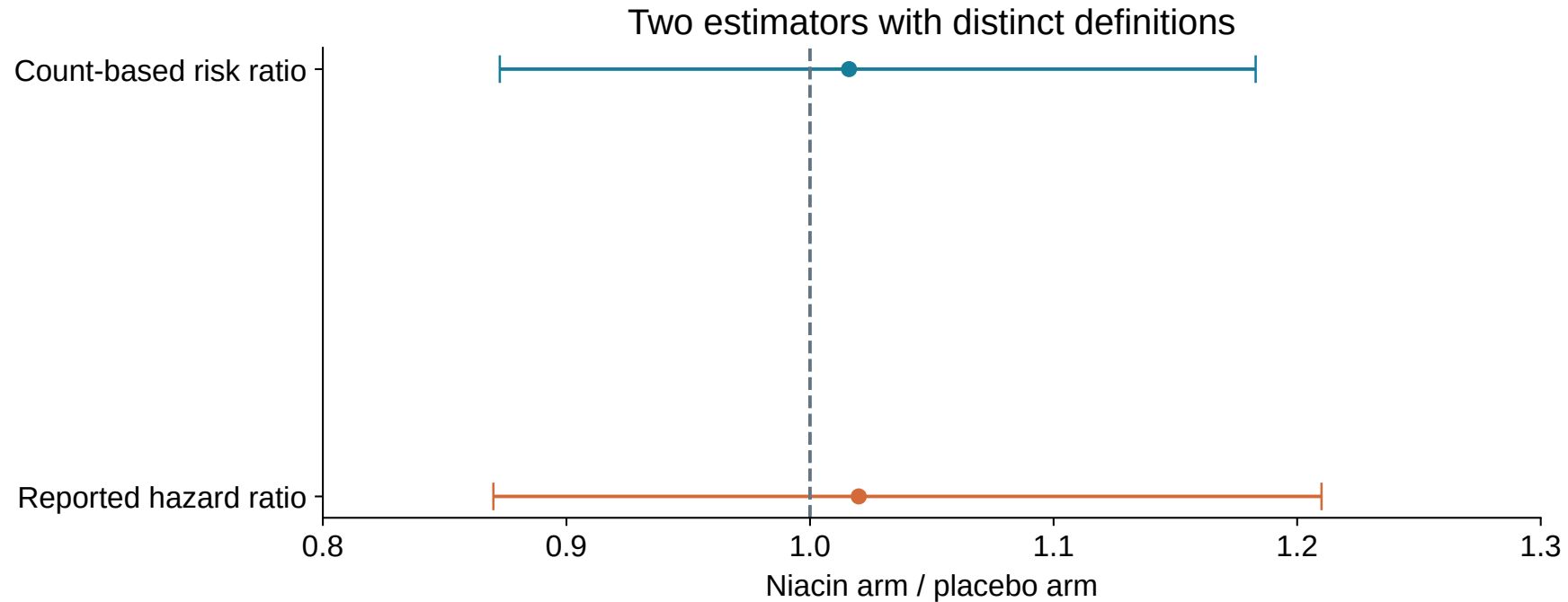
Assigned add-on	Participants n	With primary endpoint e
Niacin	1,718	282
Placebo	1,696	274

- Niacin arm risk = 16.41% Placebo arm risk = 16.16%
- Both groups received intensive statin treatment.
- The primary endpoint is a composite cardiovascular outcome.

18. The delta method for log risk

- $\log(\hat{p}) \approx \log(p) + (\hat{p} - p)/p$
- $\text{Var}(\log \hat{p}) \approx \text{Var}(\hat{p})/p^2 = (1 - p)/(np)$
- Substitute $\hat{p} = e/n$: estimated variance = $1/e - 1/n$
- Independent arms: variances add for $\log(\text{RR})$

19. A confidence interval for a risk ratio

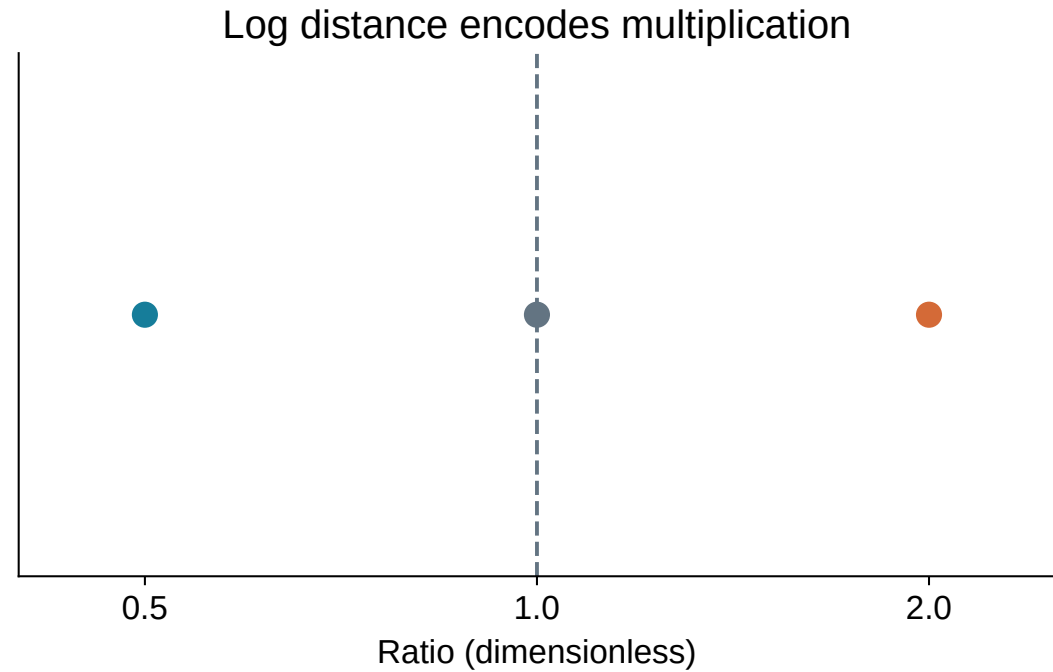
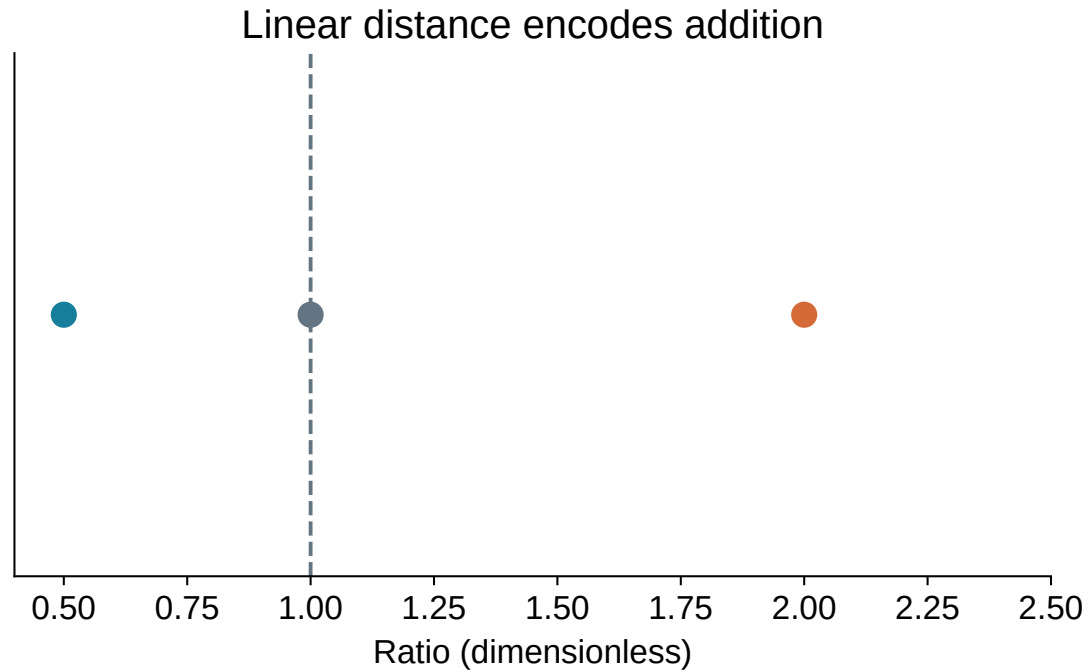


- $SE(\log RR) = \sqrt{[1/eT - 1/nT + 1/eC - 1/nC]}$
- Log interval: $\log(RR) \pm 1.96 \times SE(\log RR)$
- Exponentiating both limits returns the ratio scale.
- AIM-HIGH count-based RR: 1.016 (0.873–1.183)

20. An interval crossing 1 leaves uncertainty

- The trial did not demonstrate additional clinical benefit from the niacin add-on.
- Its reported HR interval spans 0.87–1.21.
- Compatible effects include some benefit and some harm.
- An equivalence claim requires a defined margin and suitable analysis.

21. Logarithmic ratio geometry



- $u(r) = \log(r)$ $u(1) = 0$
- $u(2) = -u(1/2)$
- $u(r_2) - u(r_1) = \log(r_2/r_1)$
- Ratio plots use reference 1. Difference plots use reference 0.

22. Checkpoint and 15-minute break

- 1. Why do bars and restricted-axis dots require different baseline decisions?
- 2. Why does the HDL rise fail to establish a treatment benefit?
- 3. Place 0.5, 1 and 2 on a log-ratio axis.
- Break: minutes 105–120

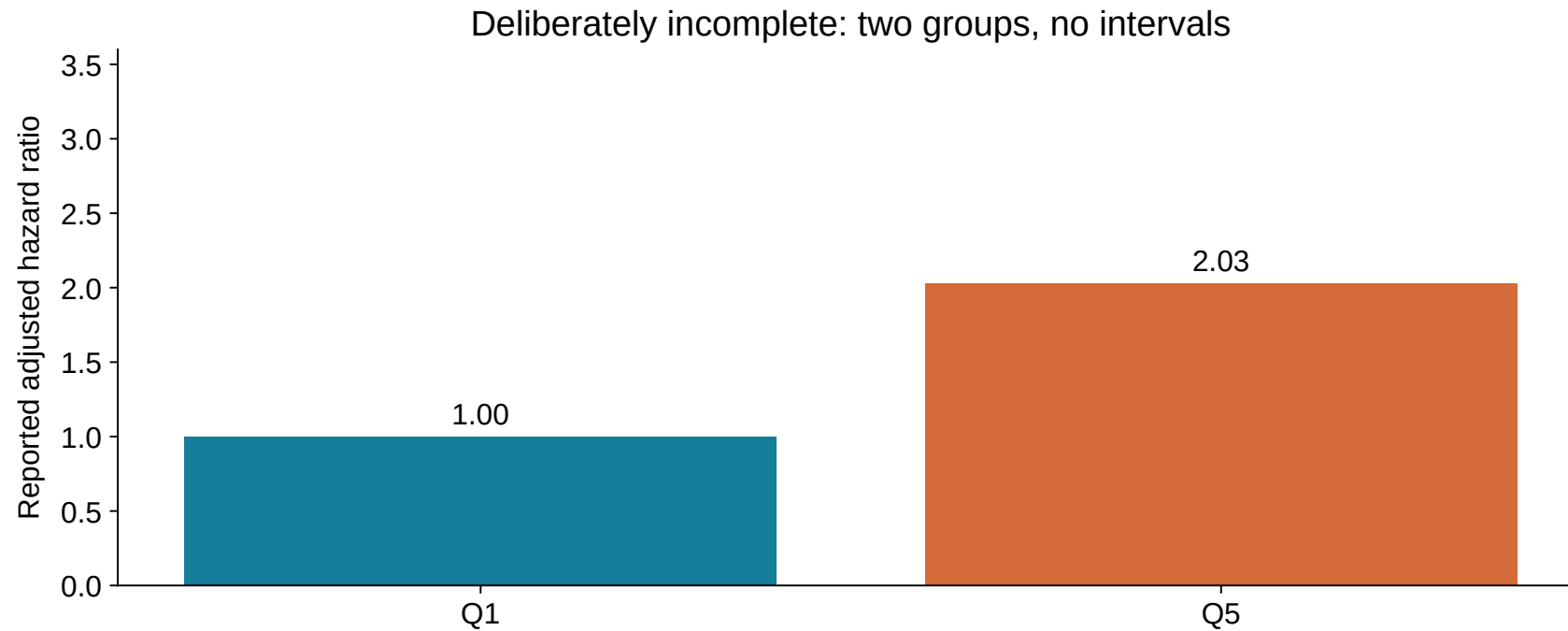
23. Added sugar as a share of energy

- $S\% = 100 \times (4 \times \text{grams of added sugar}) / \text{total kcal}$
- 50 g at 2,000 kcal = 10%
- 100 g at 4,000 kcal = 10%
- These two diets are invented arithmetic examples.

24. Cumulative risk and hazard

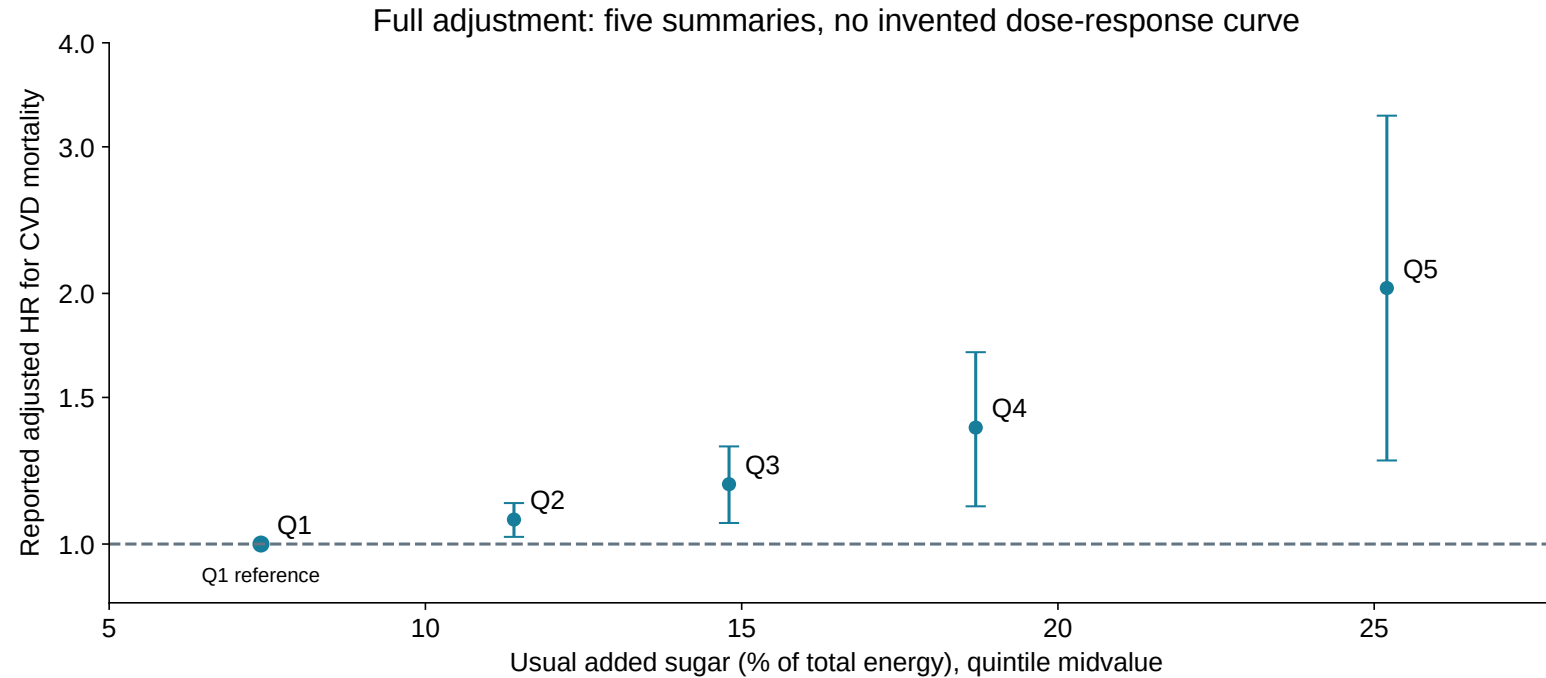
- Risk by time t : $P(T \leq t)$
- Hazard: event rate at t conditional on survival to t
- $h(t) = \lim_{\Delta t \rightarrow 0} P(t \leq T < t + \Delta t \mid T \geq t) / \Delta t$
- $HR(t) = hE(t) / hR(t)$

25. Two selected points make an incomplete pitch



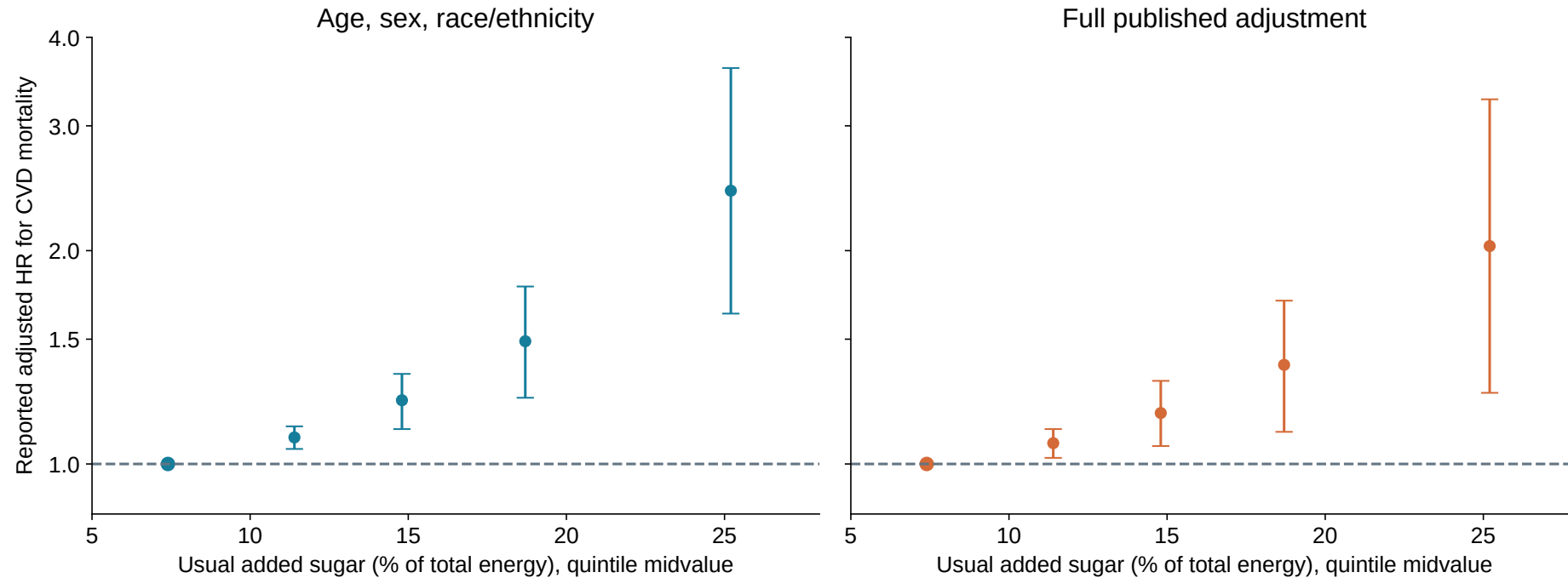
- Q5 versus Q1: reported full-adjustment HR = 2.03
- The pitch hides intermediate exposure summaries and interval width.
- Which visual omissions could turn this association into advice?

26. The full sugar evidence



- Numeric exposure positions preserve unequal spacing.
- Intervals show precision at each reported midvalue.
- Q1 is the reference. Its marker has no fabricated interval.
- Q5's 25.2% is a midvalue. Its category starts at 21.3%.

27. Both published adjustment models



- The first model adjusts for age, sex and race/ethnicity.
- The full model adds behavioral, clinical and dietary covariates.
- Full adjustment includes total serum cholesterol.
- A changed estimate alone cannot identify which covariate caused it.

28. The sugar finding's scope

- Higher modeled usual added-sugar intake remained associated with higher CVD mortality after full adjustment.
- Residual confounding and changing intake limit causal interpretation.
- Total serum cholesterol adjustment does not equal adjustment for LDL-C and HDL-C separately.
- This association does not erase the randomized treatment evidence.

29. A visualization specification

- Question: what comparison should the reader make?
- Quantity and encoding: risk difference, ratio, concentration or exposure
- Scale and reference: zero for differences, one for ratios
- Uncertainty and caption: study design, endpoint, comparison and period

30. Color, units and captions

- Direct labels preserve meaning when color disappears.
- HDL concentration and event risk need separate axes and panels.
- An interval belongs beside its estimate and null reference.
- A caption states the finding and the evidence that supports its scope.

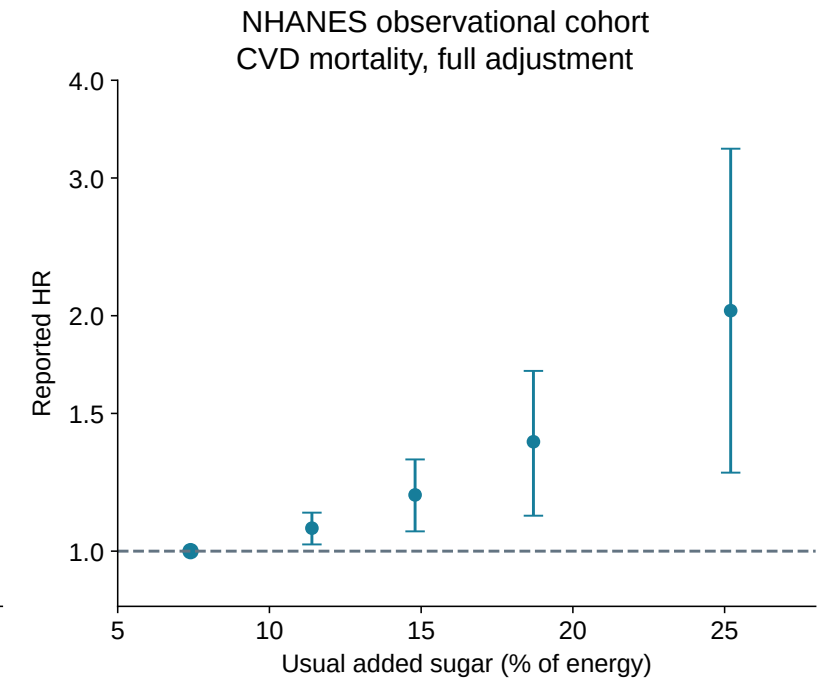
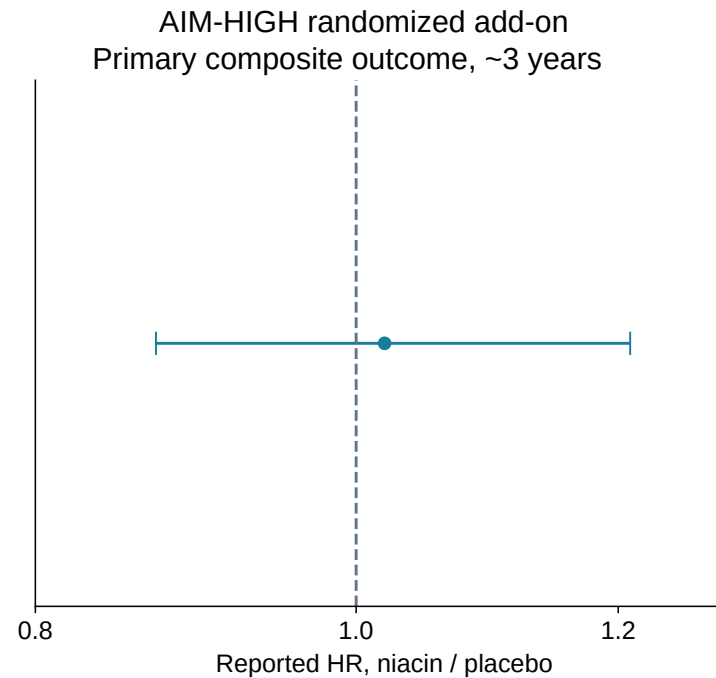
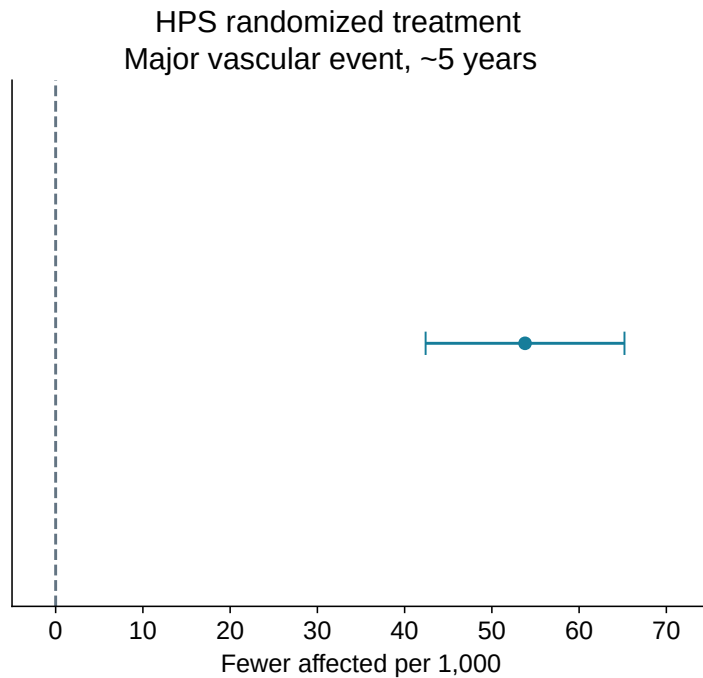
31. Evidence workshop: one pitch to repair

- 5 min: calculate an effect and identify the omission
- 7 min: sketch a repaired figure with labeled units and reference
- 5 min: write the finding, numerical evidence and study scope
- 3 min: defend the repair against a reviewer challenge

32. Reviewer challenges

- “Your figure shows a biomarker change. Where are the clinical outcomes?”
- “Your interval crosses the null. Why does the caption claim equality?”
- “Your sugar plot shows only Q1 and Q5. What did you omit?”
- “Your axes show different quantities. Why are you ranking them?”

33. Three findings with separate scopes



- HPS: treatment benefit on an absolute scale
- AIM-HIGH: reported outcome HR with its interval
- NHANES: adjusted cohort association across exposure summaries

34. The editor's publication brief

- HPS: about 54 fewer affected people per 1,000 during approximately five years.
- AIM-HIGH: the niacin add-on did not demonstrate additional clinical benefit in the studied setting.
- NHANES: the positive sugar association remained after full adjustment.
- Publish the specific results and replace the single-culprit headline.

35. Exit ticket

- 1. Translate the HPS absolute benefit into affected people per 1,000.
- 2. Explain why improved HDL does not establish protection.
- 3. Explain why the sugar association does not acquit LDL.
- 4. Give the null reference for a difference and for a ratio.

36. Sources and further reading

- HPS Collaborative Group (2002), Lancet 360:7–22
- AIM-HIGH Investigators (2011), NEJM 365:2255–2267
- Yang et al. (2014), JAMA Internal Medicine 174:516–524
- Ference et al. (2017), European Heart Journal 38:2459–2472
- Kearns et al. (2016), JAMA Internal Medicine 176:1680–1685